

ABDELRAHMAN ELSHRIEF

613-770-4883 | abdelelshrief@gmail.com | linkedin.com/in/abdel-elshrief | github.com/Abdel-E | abdele.com

Education

University of Toronto

Bachelor of Applied Science (BASC) in Computer Engineering

Expected: Apr 2028

Toronto, ON

Experience

Research Engineer Intern

May 2026 - Aug 2026

Royal Military College of Canada, NavINST Lab

Kingston, ON

- Created a lab automation tool with Python/Tkinter for GNSS jamming resilience testing allowing for waveform control, 10 dB gain stepping, receiver outage/recovery event marking, and timestamped CSV logs, improving repeatability and **reducing manual test execution and data-labeling time by an estimated 10-15 minutes per test run**.
- Developed a **low-cost 4-channel GNSS array** using a Dual-AD9361 system controlled by an external 10 MHz clock and local oscillator **allowing for synchronized multi-channel operation and Direction of Arrival estimation**.
- Evaluated **4 chirp/wideband calibration methods** and selected candidate approach using literature-reported **92-99% calibration-error reductions** over single-point techniques.
- Improved multimodal sensor alignment for NavINST's autonomous vehicle by correcting **~840 ms** of event/frame-camera timing offset, using motion-signal synchronization and digital markers to support sensor fusion and extrinsic calibration.

Machine Learning Engineering Intern

Jan 2026 - Apr 2026

Agentiiv (Agentic AI Startup)

Toronto, ON

- Built a Corporate Intelligence platform by architecting a **FastAPI MCP server** that unifies internal systems (Slack, GDrive, PostgreSQL) and monitors external data sources (social media, etc) to accelerate team processes and execution.
- Implemented SSO authentication and Role-Based-Access, allowing for scoped permissions and granular logging of usage, latency, and uptime metrics of all internal and external MCP connections using **Prometheus and Grafana**.
- Cut deployment time from **~20 to ~5 minutes** by containerizing services on AWS and automating deployment workflows using Docker Compose.

Embedded Systems Engineer

Sep 2025 - Apr 2026

University of Toronto FSAE

Toronto, ON

- Built a custom Unity/C# simulation environment with **3+ virtual sensors** (LiDAR, IMU, GPS) to support in-house **ROS 2** perception testing and reduce dependency on physical track time for validation and software development.
- Improved on-car **LiDAR ground detection by 10%** by digitally filtering noise and labeling point-cloud data.

Technical Project Manager

Jan 2026 - Present

University of Toronto Machine Intelligence Team (UTMIST)

Toronto, ON

- Selected to lead a team of **6+ engineers**, owning scope, milestones, and delivery, for the 2026/27 school year.

Projects

Cerebras Kernel Challenge - Distributed Top-K kNN | CSL, Python, WSE-2, Parallel Computing

- Implemented an **exact distributed Top-K kNN algorithm** on a SDK-simulated Cerebras WSE-2 Wafer chip grid using query broadcast, vectorized L2 distance computation, and hierarchical + heap-adjacent Top-K reduction.
- Reduced kernel cycle counts by **up to ~85%**, cutting the baseline case from **~160K to ~25K cycles** and K=256 case from **~851K to ~318K** using vector FMACs, blocked memory layouts, and K-specific optimizations.

Twin - MakeUoFT 1st Place Overall | Python, C++, Computer Vision, Raspberry Pi, Arduino

- Won **1st Place Overall among 250+ hackers** by engineering a Pi/Arduino AI wearable and reducing vision inference latency by **80%** through the analysis of real-time YOLO-v8 model labels in place of full image processing.

Conversion Rate Optimization Agent | Python, RL, Gemini API, FastAPI

- Built an autonomous storefront optimization agent to orchestrate the full user journey, from landing to checkout, **replacing static A/B testing with real-time Reinforcement Learning** using Thompson Sampling.
- Mapped visitor signals into behavioral segments to drive targeted content across landing, cart, and checkout, **outperforming traditional A/B testing by 40%** in simulations

Technical Skills

Languages: Python, C/C++, TypeScript/JavaScript, SQL

Technologies: Node.js, React, PyTorch, scikit-learn, Docker, AWS, Microcontrollers, ROS2

Engineering: CI/CD, Git, Sensor Fusion, Computer Vision, SDR, PyTest, Agile/Scrum